VANESSA MAHAN QUINTANA

Applied Ecological Modeler · Coastal Systems Scientist · Geospatial Analyst

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Applied ecological modeler developing decision-support tools for federal agencies, Tribal partners, and coastal resource managers, with 5+ years of experience in hydrodynamic and ecological modeling. Specializes in integrating hydrodynamic modeling, spatial analysis, and agent-based modeling to translate complex environmental systems into actionable insight for federal agencies, Tribal partners, and resource managers.

CORE COMPETENCIES

- Geospatial Analysis (ArcGIS / QGIS)
- Coastal & Estuarine Hydrodynamics
- Remote Sensing & LiDAR Analysis
- Ecological Modeling (Habitat/ Agent-Based)
- Python · R · NetLogo · MATLAB Scripting
- Spatial Database Management
- Automated Geoprocessing Workflows
- Federal Compliance & Technical Reporting
- Community Engagement and Co-Development

PROFESSIONAL EXPERIENCE

Ecological Modeler / ORISE Research Fellow | U.S. Army Corps of Engineers – ERDC *2021 – Present | Vicksburg, MS / Remote*

- Designed a Python automation workflow that reduced groundwater modeling runtimes from 7–14 days to under 5 minutes, enabling rapid scenario testing and on-time delivery of infrastructure decision-support analyses.
- Co-authored habitat quality and quantity assessments across six major ecosystem types in the New York Bight, producing geospatial deliverables that directly informed USACE coastal storm risk management feasibility studies.
- Led the design and implementation of a Tribal decision-support agent-based model integrating hydrodynamics, fish energetics, and migration behavior to guide river herring fishery management for the Wampanoag Tribe of Gay Head (Aquinnah).
- Developed and published HyporheicFloPy, a Python/MODFLOW automation tool for hyporheic hydraulics modeling in rivers, streamlining aquatic restoration workflows for USACE project teams.
- Co-authored ITEKVault, an AI-assisted literature synthesis tool integrating Traditional Ecological Knowledge (TEK) frameworks to support ecologically informed, community-relevant restoration planning.
- Led 45+ interdisciplinary project meetings for the WRISES group; authored 5+ technical reports and contributed to 4 manuscripts including a peer-reviewed ecological modeling paper.

PhD Researcher – Coastal & Estuarine Systems | University of Maine *2022 – Present | Orono, ME*

- Generated field-based evidence through 4+ ADCP deployments and 300+ hours of estuarine boat-based sampling; integrated 120+ water samples and 300+ GB of spatial data to characterize sediment transport and mercury exposure risk across a macrotidal estuary.
- Produced spatial risk maps identifying hotspots where hydrodynamics, suspended sediments, salinity, and fish behavior intersect to elevate contaminant exposure risk, informing coastal remediation and monitoring priorities.
- Led design of the GoFish toolkit; an agent-based modeling library for estuarine fish, incorporating tidal transport, energetics, sediment bioaccumulation, and movement ecology.
- Developed Mercury Risk Explorer, an interactive GIS decision-support platform synthesizing material transport, habitat availability, and fish migration for coastal resource managers.
- Planned and facilitated a 32-participant participatory modeling workshop with Tribal, federal, state, and academic partners, producing co-developed model objectives and behavioral rules informing the Penobscot River hydrodynamic–ABM framework.

GIS Laboratory Intern | NOAA Fisheries – Greater Atlantic Regional Fisheries Office *2021 – 2022 | New England*

- Performed spatial habitat impact analyses for major port and infrastructure projects affecting 80+ km of Atlantic sturgeon critical habitat, informing ESA Section 7 Biological Opinion determinations on $300M+ in federal projects.
- Developed scripted R and ArcGIS Pro spatial analysis workflows enabling reproducible, defensible regulatory impact assessments; identified and quantified impacts to Primary Biological Features (PBF1 and PBF2) for avoidance and mitigation planning.
- Authored NOAA's Section 7 GIS Guidance Document (Policy Series 22-03), standardizing geospatial methods and best practices for federal ESA consultations in the Greater Atlantic region.

GIS Analyst | TRC Solutions – Environmental Consulting *2020 – 2021 | Kansas City, MO*

- Maintained and updated an enterprise GIS database representing 15,000,000+ ft of pipeline infrastructure at ≤10-meter spatial accuracy, supporting environmental review, permitting, risk assessment, and emergency response planning.
- Completed 35 backlogged GIS reconstruction projects within 10 months, integrating engineering, construction, and environmental data sources to restore usable spatial records for regulatory compliance and operational decision-making.

TECHNICAL SKILLS

GIS Platforms: ArcGIS Pro, ArcMap, QGIS, ArcGIS Online, GeoPandas, sf (R), terra (R)

Programming: Python (Flopy, Pandas, NumPy, GeoPandas, Scikit-learn, TensorFlow), R (sf, raster, ggplot2, Shiny, terra), MATLAB, NetLogo

Spatial & Modeling: Hydrodynamic modeling, agent-based modeling, habitat suitability modeling, groundwater flow & particle tracking, machine learning, Bayesian modeling

Data & Reporting: ADCP data processing, water quality analysis, time-series analysis, R Shiny dashboards, Jupyter notebooks, Markdown/Bookdown, GitHub

Other: Technical report writing, stakeholder engagement, ESA compliance support, field operations, boating safety

PUBLICATIONS & TECHNICAL PRODUCTS

Peer-Reviewed & Technical Reports

Swannack, T. M., Cushway, K. C., Carrillo, C. C., Calvo, C., Determan, K. R., Mierzejewski, C. M., **Quintana, V. M.**, Riggins, C. L., Sams, M. D., & Wadsworth, W. E. (2025). Cracking the code: Linking good modeling and coding practices for new ecological modelers. *Ecological Modelling*, 499, 110926. https://doi.org/10.1016/j.ecolmodel.2024.110926

Mahan, V., ten Brink, T., & Johnsen, P. (2022). Guidance for integrating GIS analyses into ESA Section 7 federal consultations. *Greater Atlantic Region Policy Series* [22-03]. NOAA Fisheries. greateratlantic.fisheries.noaa.gov/policyseries

McKay, S. K., **Mahan, V.**, Dougherty, M. P., Swannack, T. M., Hall, C., Saltus, C. L., Reif, M. K., & Allen, S. (2022). New York Bight Ecological Model (NYBEM), Version 1.0.0. ERDC TR-EMRRP. U.S. Army Engineer Research and Development Center, Vicksburg, MS.

Menichino, G., **Quintana, V. M.**, Wiest, S., Mierzejewski, C., & Schwartz, B. (2025, *in review*). Hyporheic Exchange (HYPE) Tool: Automated hyporheic hydraulics modeling in rivers using Python. ERDC TN-ANSRP. U.S. Army Engineer Research and Development Center.

U.S. Army Corps of Engineers & Wampanoag Tribe of Gay Head (Aquinnah). (2025, *in review*). Environmental Resource Management Study in Aquinnah, Massachusetts. Final Technical Report. USACE New England District and ERDC.

United States. National Marine Fisheries Service. Greater Atlantic Regional Fisheries Office (2022). National Marine Fisheries Service Endangered Species Act Section 7 Biological Opinion USACE Permit for the New Jersey Wind Port (NAP-2019-01084-39). https://doi.org/10.25923/j8gz-g091

United States. National Marine Fisheries Service. Greater Atlantic Regional Fisheries Office (2023). USACE Permit for the Edgemoor Container Port (NAP-2019-278-23). https://doi.org/10.25923/b7zk-aa69

Software & Technical Products

Quintana, V. M. (2025). Mercury Risk Explorer [Shiny application]. vkzfr3-vanessa-mahan.shinyapps.io/Mercury_Risk_Explorer

Quintana, V. M. & Menichino, G. (2025). HyporheicFloPy: Automated hyporheic hydraulics modeling in rivers using Python [Software]. usace-wrises.github.io/HyporheicFloPy

Quintana, V. M. (2025). GoFish: A next-generation toolkit for modeling migratory fish and quantifying environmental risk in estuaries [Software documentation]. vmahan1998.github.io/GoFish

Quintana, V. M. (2024). River Herring Habitat Model [Habitat suitability model]. github.com/vmahan1998/RiverHerringHabitatModel_TechReport

Quintana, V. M. (2024). River Herring Ecological Modeling Dashboard [Shiny application]. vkzfr3-vanessa-mahan.shinyapps.io/rh_dashboard

EDUCATION

PhD Candidate · Marine Biology · University of Maine *(2022 – Present)*
Focus: Coastal Hydrodynamics, Ecological Risk Modeling, Spatial Analysis

M.S. · Environmental Science and Policy · Johns Hopkins University *(2021)*
Focus: Spatial Analysis, Landscape Ecology | Certificate in GIS Applications

B.S. · Biological Sciences · Missouri University of Science & Technology *(2020)*
Minor in Chemistry | Certificate in Bio-Innovation

HONORS & RECOGNITION

Graduate Climate Conference Poster Award · MIT / Woods Hole Oceanographic Institution (2025) | **ORISE Research Fellowship** · Oak Ridge Institute (2021–2026) | **Forbes 30 Under 30 Healthcare Nominee** · 2020 | COPRI Student Award & CERF Travel Grant (2025) | Entrepreneur Quest 3rd Place ($5,000), 2nd Place ($10,000) · 2019, 2020

PROFESSIONAL AFFILIATIONS & LEADERSHIP

COPRI (ASCE) Student Chapter President (2025–Present) — Rebuilt chapter into interdisciplinary hub for coastal engineering and coastal science students | **CERF · AFS · AAAS** — Active member; presented research and contributed to working groups on co-development, ecological modeling, and coastal resilience